Supplementary Methods and Results for "Distributionally Robust Learning for Social Program Participation Prediction: A Cross-Department Evaluation in Peru"

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Scope. This supplement reports variable definitions, training specifications, departmental prediction estimates, calibration analyses and sensitivity results from ENAPRES 2025. Tables and figures use supplementary numbering. All results refer to household-reported Pensión 65 and Juntos participation in the two age-restricted samples.

1 Records, outcomes and predictor audit

1.1 Unit of analysis and analytic restrictions

The 2025 National Survey of Budget Programmes (ENAPRES 2025) contributed 36,184 households from 24 departments and Callao. Unique records were constructed from year, month, department, conglomerate, dwelling and household codes. A positive programme indicator denotes reported benefit receipt by at least one member of the household. Geographic and household identifiers served only to define records or evaluation groups.

The Pensión 65 analytic subset contained 11,354 households with at least one older adult (65 years or more), including 4,531 positive records. The Juntos subset contained 17,671 households with at least one member aged 0–17 years, including 3,341 positive records. Weighted prevalence was 26.86% and 14.90%, respectively. The programmes represent old-age income support and conditional household transfers associated with health and education, respectively [8,7]. The same 25-department validation design was applied separately to their distinct populations. In the all-household Juntos sensitivity sample, 3,358 of 36,184 records were positive, 17 more than in the age-restricted task. The two primary subsets overlap.

The household survey weights summed to approximately 3.09 million for Pensión 65 and 4.06 million for Juntos. Kish weight-only effective sizes were 4,732 and 7,365, respectively [3]. Approximate primary sampling units combined department and conglomerate identifiers while preserving leading zeros; the subsets

contained 3,810 and 4,262 units. Variance calculations use these approximate units [5].

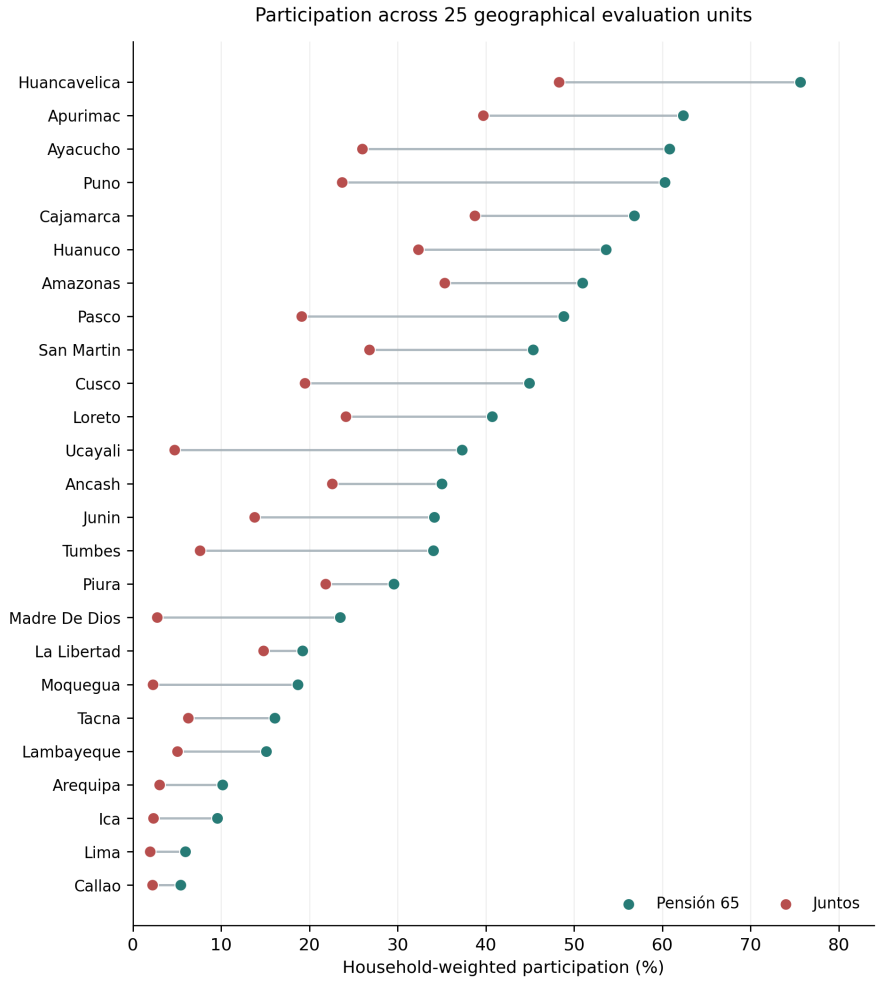


Fig. S1. Territorial variation in weighted household participation in the two distinct age-restricted task samples. Connecting lines join the two task-specific prevalences for each department; their denominators are different. Unweighted support and full scores appear in Tables S3–S4.

1.2 Predictor coding and missingness

The primary models used 18 predictors: 14 numeric or binary and four categorical variables (Table S1). Disability was represented by an indicator of whether any household member reported a disability. The asset index was derived from

17 household asset indicators separately within each training partition. Source column names and their analytic meanings appear below.

Table S1. Source field identifiers and analytical definitions for the 18 household predictors.

Group	Source column	Recorded meaning
Demographic	`edad_jefe`	Age of household head (years)
Demographic	`tamano_hogar`	Household size
Demographic	`n_ninos_0_11`	Number of children aged 0–11
Demographic	`n_adolescentes_12_17`	Number of adolescents aged 12–17
Demographic	`n_adultos_65mas`	Number of adults aged 65 years or older
Demographic	`n_mujeres`	Number of female household members
Housing	`agua_red_publica`	Connection to public piped-water supply (0/1)
Housing	`saneamiento_red_publica`	Connection to public sewerage network (0/1)
Housing	`electricidad_red_publica`	Grid electricity access (0/1)
Household	`hogar_multigeneracional`	Multigenerational household (0/1)
Household	`hogar_con_discapacidad`	At least one household member with a disability (0/1)
Household	`indice_activos_pca`	Training-fold first principal component of 17 household asset indicators, standardized by source weights
Housing	`internet_en_hogar`	Household internet access (0/1)
Housing	`hogar_con_dispositivo_digital`	Household owns/has a digital device (0/1)
Categorical	`sexo_jefe`	Gender of household head (male/female)
Categorical	`educacion_jefe_4niveles`	Household head education (four codes)
Categorical	`lengua_materna_jefe`	Mother tongue of household head (coded)
Categorical	`area`	Urban (code 1) or rural (code 2); source code retained

Education and mother tongue entered as categorical predictors. Logistic regression and neural models used one-hot encoding fitted within training departments; unseen categories at evaluation were represented by all-zero indicator blocks. Numerical standardization used source-household weights. CatBoost used its native categorical encoding [9].

The evaluation group was defined by `codigo_departamento`. Household identifiers comprised `ANIO`, `MES`, `CONGLOMERADO`, `VIVIENDA` and `HOGAR`; none was

included as a predictor. Other programme indicators were excluded. Natural region (`region_natural`) was assessed in a fixed-specification sensitivity analysis. Indigenous self-identification had 2,361 missing values in the household file and was assessed in a complete-case analysis. All 18 primary predictors were available in the two analytic populations.

1.3 Training-partition principal component of household assets

For each outer held-out department, the first asset principal component was obtained from the remaining 24 departments, using the 17 binary ownership indicators. Within each training partition, every indicator was centered and divided by its survey-weighted standard deviation. The first eigenvector was extracted from the weighted correlation matrix and oriented toward increasing total asset ownership. Scores were then standardized to mean zero and standard deviation one under the source-household weights. The same centering, eigenvector, sign and scaling were applied to the excluded department. An independent component fit was performed within each inner training split during hyperparameter selection.

Across the 25 outer partitions, the first component explained between 34.7% and 35.6% of weighted asset variation for Pensión 65 and between 27.8% and 28.6% for Juntos (Table S2). These ranges describe variation in the principal component fitted to different source populations rather than prediction uncertainty.

Table S2. Variation in the first survey-weighted asset principal component across 25 outer training partitions.

Programme	Assets	Minimum (%)	Mean (%)	Maximum (%)
Pensión 65	17	34.7	35.3	35.6
Juntos	17	27.8	28.2	28.6

2 Prediction and validation algorithm

2.1 Outer and inner held-out units

Each outer fold excluded one entire department and trained on the remaining 24. The 25 held-out prediction sets were subsequently pooled, with each household appearing once in the out-of-department evaluation. The same predictions were used in all reported subgroup analyses.

Within the 24 development departments, two `GroupKFold` validation splits held out complete source departments. Logistic regression, CatBoost and ERM candidates were ranked by pooled weighted validation log-loss. Group DRO candidates were ranked by the maximum weighted loss across the 24 source validation departments. Numeric scaling, categorical encoding and the asset principal component were refitted in each inner source training partition [11].

$$\widehat{R}_{d,m} = \frac{\sum_{i \in d} w_i \ell(y_i, \widehat{p}_i^{(-d,m)})}{\sum_{i \in d} w_i}, \qquad \ell(y,p) = -y \log p - (1-y)\log(1-p). \quad (1)$$

Training weights were scaled to mean one; the reported losses used the original factors. Probabilities were clipped numerically before logarithms were calculated. The source-prevalence baseline yielded one constant per excluded department. Consequently, its pooled ROC AUC reflects the ranking of fold-specific constants, whereas within-department ROC AUC equals 0.5.

2.2 Model specifications

Logistic regression used $C \in \{0.1, 1, 10\}$ and a quasi-Newton optimizer. CatBoost compared depths four and six using 240 boosting iterations, learning rate 0.05, ℓ_2 leaf penalty six, Bernoulli subsampling probability 0.8, and random-strength parameter 0.5. Random seeds varied deterministically with the outer department. Model selection used only source-department validation data.

The multilayer perceptron (MLP) had two hidden layers (48 and 24 units), rectified linear activation, and one logistic output. All models used full-batch AdamW for 95 epochs, learning rate 0.004, and gradient clipping at norm five. ERM weight decay was selected from $\{0.002, 0.02\}$. Equal-department training inherited the selected decay. Group DRO jointly selected weight decay from $\{0.002, 0.02\}$ and the exponentiated-gradient step from $\{0.04, 0.12\}$ against the maximum source validation-department loss. The earlier pooled-validation selection was retained as an additional sensitivity. Probabilities were averaged across three predetermined seeds, matched between objectives and translated deterministically by outer department.

The weighted household ERM objective is

$$L_{\mathrm{ERM}}(\theta) = \frac{\sum_{i \in \mathcal{S}} w_i \ell(y_i, p_\theta(x_i))}{\sum_{i \in \mathcal{S}} w_i}. \quad (2)$$

For source department d, let $R_d(\theta) = \sum_{i \in d} w_i \ell_i / \sum_{i \in d} w_i$. Equal-domain training minimizes $D^{-1} \sum_d R_d(\theta)$, where $D = 24$. In Group DRO, adversarial weights are initialized uniformly and updated once per training epoch,

$$q_d^{t+1} = \frac{q_d^t \exp[\eta\{R_d(\theta_t) - \max_h R_h(\theta_t)\}]}{\sum_h q_h^t \exp[\eta\{R_h(\theta_t) - \max_j R_j(\theta_t)\}]}, \qquad L_{\mathrm{DRO}}(\theta_t) = \sum_d q_d^{t+1} R_d(\theta_t). \quad (3)$$

Subtracting the maximum source risk in the exponentiated update improves numerical stability while leaving normalized adversarial weights unchanged. Weights q_d were detached when differentiating the network objective with respect to θ. This implementation follows a restricted group-robust training configuration [12].

3 Loss aggregation and uncertainty

Pooled log-loss was computed over all held-out household predictions using their expansion factors; macro-department loss was the unweighted mean of 25 department-specific losses. We also computed the maximum, the mean of the five largest departmental losses, and the empirical 90th percentile (Table S5). These statistics summarize different portions of the departmental risk distribution.

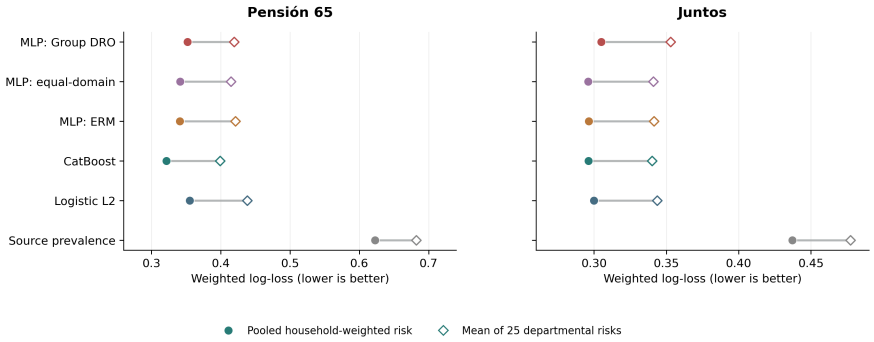


Fig. S2. Pooled factor-weighted and equal-department-averaged log-loss computed from the same held-out predictions. Connecting segments indicate the aggregation contrast.

$$H_d = -\widehat{\pi}_d \log \widehat{\pi}_d - (1-\widehat{\pi}_d)\log(1-\widehat{\pi}_d), \qquad Q_m = \max_d \frac{\widehat{R}_{d,m}}{H_d}. \tag{4}$$

The denominator H_d uses held-out departmental prevalence and represents the constant-probability benchmark fitted to the same department. The statistic Q_m is evaluated after observing the test outcomes and complements the absolute-loss comparison.

For pooled contrasts, 1,200 resamples drew approximate clusters with replacement within departments, keeping the original paired prediction sets. For maximum-loss contrasts, 1,600 analogous resamples recomputed the maximum across departments. Departmental calibration used 1,200 within-department grouped resamples. The eight pooled tests were adjusted using Benjamini–Hochberg [1]. Resampling varied the grouped observations while retaining model parameters and hyperparameter selections.

4 Complete departmental predictive losses

Tables S3 and S4 report all 25 held-out departmental losses for each model. The column p_d denotes weighted participation in the corresponding age-restricted task sample.

Table S3. Complete department-held-out weighted log-loss for Pensión 65. The last five columns come from separate models trained without that department.

Department	p_d (%)	LR	Cat	ERM	Equal	DRO
Amazonas	50.9	0.618	0.548	0.594	0.567	0.540
Ancash	35.0	0.346	0.309	0.316	0.326	0.337
Apurimac	62.3	0.409	0.347	0.391	0.375	0.375
Arequipa	10.2	0.229	0.207	0.217	0.228	0.240
Ayacucho	60.8	0.504	0.440	0.471	0.449	0.444
Cajamarca	56.8	0.575	0.461	0.514	0.497	0.496
Callao	5.4	0.208	0.196	0.201	0.211	0.234
Cusco	44.9	0.408	0.357	0.379	0.379	0.386
Huancavelica	75.7	0.482	0.398	0.431	0.416	0.413
Huanuco	53.6	0.475	0.419	0.436	0.434	0.444
Ica	9.5	0.307	0.284	0.292	0.312	0.329
Junin	34.2	0.484	0.432	0.468	0.469	0.479
La Libertad	19.2	0.330	0.312	0.323	0.341	0.354
Lambayeque	15.1	0.340	0.323	0.339	0.359	0.399
Lima	6.0	0.197	0.182	0.186	0.190	0.200
Loreto	40.7	0.511	0.469	0.505	0.471	0.458
Madre De Dios	23.5	0.438	0.374	0.393	0.396	0.401
Moquegua	18.7	0.306	0.302	0.299	0.314	0.347
Pasco	48.8	0.580	0.534	0.553	0.527	0.525
Piura	29.5	0.424	0.399	0.430	0.428	0.451
Puno	60.3	0.530	0.475	0.554	0.565	0.612
San Martin	45.4	0.673	0.655	0.664	0.607	0.550
Tacna	16.1	0.354	0.336	0.349	0.350	0.362
Tumbes	34.1	0.640	0.650	0.672	0.637	0.616
Ucayali	37.3	0.587	0.566	0.549	0.524	0.499

Table S4. Complete department-held-out weighted log-loss for Juntos. The last five columns come from separate models trained without that department.

Department	p_d (%)	LR	Cat	ERM	Equal	DRO
Amazonas	35.3	0.632	0.621	0.632	0.638	0.587
Ancash	22.6	0.323	0.329	0.335	0.336	0.366
Apurimac	39.7	0.528	0.519	0.523	0.514	0.481
Arequipa	3.0	0.135	0.129	0.134	0.135	0.166
Ayacucho	26.0	0.437	0.437	0.440	0.439	0.469
Cajamarca	38.8	0.559	0.556	0.559	0.564	0.548
Callao	2.2	0.115	0.114	0.110	0.110	0.131
Cusco	19.5	0.363	0.361	0.358	0.362	0.391
Huancavelica	48.3	0.650	0.638	0.647	0.636	0.601
Huanuco	32.4	0.493	0.490	0.484	0.480	0.504
Ica	2.3	0.118	0.118	0.118	0.122	0.150
Junin	13.8	0.300	0.297	0.293	0.295	0.327
La Libertad	14.8	0.314	0.306	0.310	0.307	0.322
Lambayeque	5.0	0.194	0.191	0.195	0.194	0.246
Lima	1.9	0.105	0.100	0.098	0.097	0.106
Loreto	24.2	0.474	0.457	0.456	0.462	0.455
Madre De Dios	2.8	0.146	0.141	0.143	0.151	0.207
Moquegua	2.3	0.138	0.140	0.140	0.143	0.178
Pasco	19.1	0.439	0.442	0.448	0.442	0.451
Piura	21.8	0.451	0.465	0.458	0.448	0.415
Puno	23.7	0.463	0.449	0.461	0.463	0.447
San Martin	26.8	0.518	0.510	0.508	0.507	0.479
Tacna	6.2	0.189	0.185	0.187	0.179	0.196
Tumbes	7.6	0.254	0.267	0.252	0.252	0.275
Ucayali	4.7	0.254	0.240	0.245	0.252	0.322

For Pensión 65, CatBoost's highest departmental loss was 0.655 in San Martin; its lowest was 0.182 in Lima. For Juntos, the corresponding extremes were 0.638 in Huancavelica and 0.100 in Lima. These losses are measured on the same logarithmic scale but reflect the outcome distributions of different analytic populations.

Table S5. Additional risk summaries from the same 25 held-out domains: mean of the five largest losses, 90th percentile of departmental losses, and maximum loss divided by local outcome entropy.

Task	Model	Pooled	Macro	Max	Top-5	P90	Max/H
P65	Logistic L2	0.3551	0.4382	0.6734	0.6198	0.6060	0.997
P65	CatBoost	0.3217	0.3990	0.6548	0.5906	0.5588	1.014
P65	MLP–ERM	0.3412	0.4211	0.6715	0.6073	0.5777	1.047
P65	MLP–equal	0.3414	0.4149	0.6374	0.5807	0.5660	1.004
P65	MLP–DRO	0.3520	0.4196	0.6161	0.5685	0.5460	1.113
Juntos	Logistic L2	0.3000	0.3436	0.6498	0.5775	0.5467	1.336
Juntos	CatBoost	0.2962	0.3402	0.6384	0.5689	0.5415	1.288
Juntos	MLP–ERM	0.2965	0.3414	0.6466	0.5738	0.5448	1.289
Juntos	MLP–equal	0.2959	0.3411	0.6379	0.5719	0.5440	1.328
Juntos	MLP–DRO	0.3050	0.3528	0.6010	0.5441	0.5302	1.695

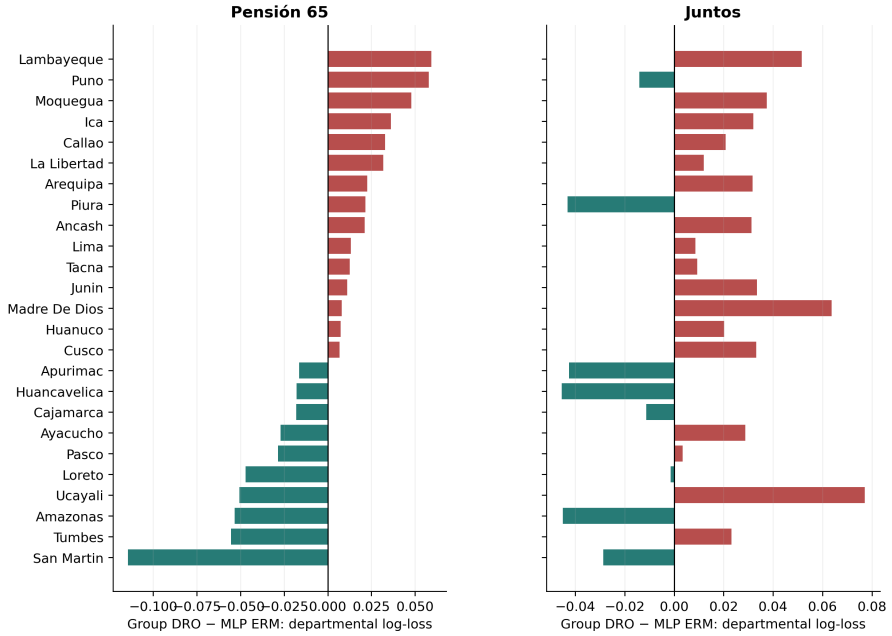


Fig. S3. Difference between Group DRO and neural ERM departmental weighted log-loss. Negative values favour Group DRO.

With worst-department selection, Group DRO achieved lower loss than neural ERM in 10 of 25 departments for Pensión 65 and 8 for Juntos. Maximum losses under Group DRO and ERM were 0.6161 and 0.6715 for Pensión 65, and 0.6010 and 0.6466 for Juntos. Maximum loss-to-local-entropy ratios were 1.113 and 1.047, and 1.695 and 1.289, respectively. Mean losses across the five highest-risk departments were 0.5685 versus 0.6073 and 0.5441 versus 0.5738.

5 Paired predictive comparisons and worst-risk intervals

Table S6. Exploratory paired comparison of equal-department neural training minus CatBoost in pooled household-weighted log-loss. Percentile intervals use approximate clusters sampled within departments, conditional on fixed predictions; no multiplicity adjustment was applied to this supplementary contrast.

Task	Δ	95% interval	Approx. p	Resamples
P65	+0.01975	[+0.01512, +0.02416]	0.000	2,000
Juntos	-0.00024	[-0.00211, +0.00169]	0.799	2,000

The paired equal-department neural minus CatBoost difference for Juntos is an exploratory comparison separate from the original eight-contrast testing family. The interval is conditional on out-of-department predictions and the approximate

clustering of survey records. The point difference does not establish superiority of either specification.

Table S7. All eight prespecified national log-loss contrasts. Percentile ranges derive from 1,200 stratified-by-department approximate-cluster reweightings of fixed predictions; the final column is BH-adjusted across these eight comparisons.

Task	First − second	Δ	95% interval	BH q
P65	CatBoost - logistic	-0.03342	[-0.03891, -0.02797]	6.425e-32
P65	CatBoost - ERM	-0.01950	[-0.02340, -0.01545]	3.655e-21
P65	DRO - ERM	+0.01076	[+0.00601, +0.01572]	3.558e-05
P65	Equal - ERM	+0.00025	[-0.00267, +0.00302]	0.8642
Juntos	CatBoost - logistic	-0.00378	[-0.00565, -0.00177]	0.0001785
Juntos	CatBoost - ERM	-0.00029	[-0.00209, +0.00143]	0.8496
Juntos	DRO - ERM	+0.00852	[+0.00421, +0.01284]	0.0001158
Juntos	Equal - ERM	-0.00053	[-0.00177, +0.00064]	0.5236

CatBoost–logistic pooled log-loss differences were -0.0334 for Pensión 65 and -0.0038 for Juntos. CatBoost–neural ERM differences were -0.0195 and -0.0003, respectively. The eight comparisons, percentile intervals and BH-adjusted q values are reported in Table S7.

Table S8. Exploratory differences in maximum observed departmental log-loss. The bootstrap recalculates the maximum after each of 1,600 resamples; it never retrains the models. These are nonlinear functions of the 25 included departments, not predictions for an unseen domain distribution.

Task	First − second	$\Delta_{\max}$	95% interval	SE
P65	DRO - ERM	-0.0554	[-0.1569, +0.0165]	0.0440
P65	Equal - ERM	-0.0341	[-0.0823, -0.0162]	0.0177
P65	CatBoost - DRO	+0.0387	[-0.0336, +0.1458]	0.0455
P65	CatBoost - ERM	-0.0167	[-0.0471, +0.0160]	0.0161
Juntos	DRO - ERM	-0.0455	[-0.0691, -0.0280]	0.0103
Juntos	Equal - ERM	-0.0087	[-0.0162, +0.0108]	0.0085
Juntos	CatBoost - DRO	+0.0373	[+0.0135, +0.0673]	0.0133
Juntos	CatBoost - ERM	-0.0082	[-0.0234, +0.0043]	0.0070

In each grouped resample, the maximum was recalculated across the 25 weighted departmental risks. The Group DRO–ERM differences were -0.0554 (95% interval [-0.1569, 0.0165]) for Pensión 65 and -0.0455 (95% interval [-0.0691, -0.0280]) for Juntos.

6 Probability calibration

Calibration-in-the-large for evaluation set A was calculated as $\sum_{i\in A} w_i(\hat{p}_i - y_i)/\sum_{i\in A} w_i$, where a negative sign denotes underprediction. Pooled calibration slopes and intercepts were estimated using weighted logistic recalibration; reliability plots used ten bins with approximately equal factor-weighted mass. Departmental percentile intervals were obtained by resampling approximate clusters [2,13].

Table S9. Model-level held-out calibration diagnostics. ECE uses ten equally weighted-mass bins; a and b are pooled calibration intercept and slope.

Programme	Model	ECE	Bias (pp)	a	b
P65	Logistic L2	0.006	+0.04	+0.006	1.018
P65	CatBoost	0.010	+0.04	+0.015	1.036
P65	MLP: ERM	0.013	-0.16	-0.004	0.958
P65	MLP: equal-domain	0.019	+1.78	-0.160	1.005
P65	MLP: Group DRO	0.037	+3.66	-0.315	1.024
Juntos	Logistic L2	0.011	-0.11	+0.000	0.990
Juntos	CatBoost	0.007	-0.21	+0.023	1.000
Juntos	MLP: ERM	0.006	-0.37	-0.005	0.956
Juntos	MLP: equal-domain	0.006	-0.18	-0.006	0.976
Juntos	MLP: Group DRO	0.045	+4.46	-0.428	1.023

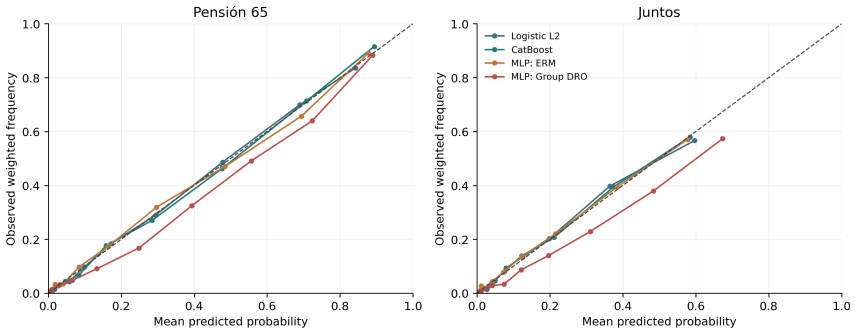


Fig. S4. Ten-bin weighted reliability plots for both outcomes and prediction methods; bins contain approximately equal weighted household mass.

6.1 Pension 65: departmental calibration

Table S10. Department-level calibration-in-the-large for Pensión 65. The interval uses 1,200 approximate-cluster replicates; predictions are held fixed.

Department	n	Observed (%)	Predicted (%)	Bias (pp)	95% range (pp)
Amazonas	462	50.9	39.5	-11.4	[-16.4, -6.8]
Ancash	483	35.0	35.9	+0.9	[-2.4, +4.4]
Apurimac	426	62.3	55.7	-6.6	[-9.7, -3.4]
Arequipa	483	10.2	14.9	+4.8	[+1.7, +7.5]
Ayacucho	493	60.8	52.9	-7.9	[-12.2, -3.8]
Cajamarca	411	56.8	52.2	-4.6	[-9.5, +0.3]
Callao	459	5.4	7.8	+2.4	[+0.4, +4.4]
Cusco	486	44.9	45.2	+0.2	[-3.7, +4.1]
Huancavelica	436	75.7	68.3	-7.4	[-11.8, -2.9]
Huanuco	476	53.6	53.3	-0.3	[-3.6, +3.3]
Ica	345	9.5	14.2	+4.7	[-0.1, +9.3]
Junin	497	34.2	35.4	+1.3	[-2.4, +4.9]
La Libertad	496	19.2	22.0	+2.8	[-0.4, +5.8]
Lambayeque	511	15.1	19.8	+4.7	[+0.2, +9.1]
Lima	1063	6.0	10.1	+4.1	[+2.7, +5.5]
Loreto	399	40.7	27.5	-13.2	[-17.0, -9.2]
Madre De Dios	195	23.5	22.9	-0.6	[-6.6, +5.5]
Moquegua	441	18.7	27.8	+9.1	[+6.3, +11.8]
Pasco	374	48.8	40.0	-8.8	[-14.1, -3.1]
Piura	554	29.5	26.7	-2.8	[-6.5, +0.6]
Puno	505	60.3	54.9	-5.4	[-9.8, -1.2]
San Martin	441	45.4	28.3	-17.0	[-23.4, -10.4]
Tacna	294	16.1	23.6	+7.5	[+2.5, +12.5]
Tumbes	342	34.1	22.4	-11.6	[-20.1, -3.8]
Ucayali	282	37.3	25.3	-12.0	[-18.5, -6.0]

CatBoost's greatest negative departmental calibration bias was -17.0 percentage points in San Martin (95% grouped-resampling interval [-23.4, -10.4]); its greatest positive bias was 9.1 points in Moquegua. The department-specific estimates and intervals are reported in Table S10.

6.2 Juntos: departmental calibration

Table S11. Department-level calibration-in-the-large for Juntos. The interval uses 1,200 approximate-cluster replicates; predictions are held fixed.

Department	n	Observed (%)	Predicted (%)	Bias (pp)	95% range (pp)
Amazonas	798	35.3	22.9	-12.4	[-15.9, -8.8]
Ancash	635	22.6	24.3	+1.7	[-2.0, +5.9]
Apurimac	651	39.7	30.4	-9.2	[-13.5, -5.3]
Arequipa	610	3.0	6.2	+3.2	[+1.4, +4.7]
Ayacucho	725	26.0	27.3	+1.3	[-1.5, +4.4]
Cajamarca	578	38.8	28.9	-9.8	[-14.8, -4.5]
Callao	651	2.2	3.8	+1.6	[+0.5, +2.6]
Cusco	608	19.5	22.5	+3.0	[-0.5, +6.5]
Huancavelica	586	48.3	35.1	-13.1	[-18.8, -7.2]
Huanuco	704	32.4	34.1	+1.8	[-2.7, +5.5]
Ica	555	2.3	5.5	+3.1	[+1.6, +4.5]
Junin	680	13.8	16.9	+3.1	[+0.5, +6.0]
La Libertad	777	14.8	14.6	-0.2	[-2.9, +2.6]
Lambayeque	783	5.0	12.4	+7.4	[+5.0, +9.7]
Lima	1323	1.9	5.0	+3.0	[+2.1, +3.9]
Loreto	1024	24.2	19.7	-4.4	[-8.4, -0.7]
Madre De Dios	604	2.8	9.7	+7.0	[+5.5, +8.4]
Moquegua	494	2.3	9.0	+6.7	[+5.0, +8.7]
Pasco	565	19.1	17.7	-1.4	[-4.5, +1.8]
Piura	934	21.8	13.6	-8.2	[-11.9, -4.5]
Puno	573	23.7	18.6	-5.1	[-9.3, -0.5]
San Martin	949	26.8	18.3	-8.5	[-13.0, -3.9]
Tacna	425	6.2	8.7	+2.5	[-1.2, +5.7]
Tumbes	595	7.6	8.6	+1.0	[-2.3, +4.1]
Ucayali	844	4.7	16.4	+11.7	[+9.3, +14.3]

CatBoost's greatest negative departmental calibration bias was -13.1 percentage points in Huancavelica (95% grouped-resampling interval [-18.8, -7.2]); its greatest positive bias was 11.7 points in Ucayali. The department-specific estimates and intervals are reported in Table S11.

7 Demographic and territorial subsets

Tables S12 and S13 give log-loss and calibration estimates in residence, natural-region and age-composition subsets. The analyses rescore the original held-out predictions. Residence and natural region provide different, overlapping partitions of the observations.

Table S12. CatBoost source-only LODO predictions evaluated in prespecified, potentially overlapping household subsets for Pensión 65. Rows partition the task sample only within their own grouping.

Household group	n	Prevalence (%)	Log-loss	AUC	Bias (pp)
Urban	6,463	13.5	0.2821	0.865	+0.41
Rural	4,891	70.8	0.4518	0.824	-1.19
Coast	4,154	9.1	0.2355	0.855	+3.48
Highlands	5,132	50.2	0.3949	0.902	-2.41
Amazon	2,068	45.1	0.5490	0.832	-10.71
Older adults: 1	7,947	25.3	0.3261	0.906	+0.25
Older adults: 2+	3,407	30.4	0.3117	0.928	-0.47

Table S13. CatBoost source-only LODO predictions evaluated in prespecified, potentially overlapping household subsets for Juntos. Rows partition the task sample only within their own grouping.

Household group	n	Prevalence (%)	Log-loss	AUC	Bias (pp)
Urban	11,098	6.5	0.1996	0.810	-0.04
Rural	6,573	44.7	0.6380	0.672	-0.83
Coast	6,230	4.9	0.1671	0.806	+1.72
Highlands	6,651	26.8	0.4238	0.838	-2.22
Amazon	4,790	22.7	0.4505	0.782	-2.34
Older adults: 0	15,460	15.8	0.3093	0.864	-0.26
Older adults: 1	1,699	10.2	0.2345	0.864	-0.31
Older adults: 2+	512	7.6	0.1753	0.919	+1.36

For Pensión 65, CatBoost log-loss was 0.2821 in urban and 0.4518 in rural households; the corresponding weighted participation proportions were 13.5% and 70.8%. For Juntos, urban and rural log-loss were 0.1996 and 0.6380, respectively. The residence rows summarize distinct components of each task sample, while the natural-region rows provide a separate overlapping partition.

8 Seed-level behaviour of neural models

The primary neural results used probabilities averaged across three predetermined seeds. Table S14 reports individual-seed losses. Because logarithmic loss is nonlinear, loss computed from averaged probabilities differs from the average loss of the three fitted models.

Table S14. Individual network-seed evaluations. The main manuscript uses the mean of three predicted probabilities for each observation, not the arithmetic mean of the three log-loss scores.

Task	Objective	Seed	Log-loss	Brier	Bias (pp)
P65	MLP–ERM	13	0.3426	0.1087	-0.13
P65	MLP–ERM	31	0.3424	0.1085	-0.18
P65	MLP–ERM	59	0.3425	0.1088	-0.17
P65	MLP–equal	13	0.3430	0.1088	+1.91
P65	MLP–equal	31	0.3439	0.1094	+1.68
P65	MLP–equal	59	0.3417	0.1083	+1.75
P65	MLP–DRO	13	0.3534	0.1129	+3.74
P65	MLP–DRO	31	0.3546	0.1135	+3.60
P65	MLP–DRO	59	0.3531	0.1127	+3.64
Juntos	MLP–ERM	13	0.2967	0.0929	-0.37
Juntos	MLP–ERM	31	0.2975	0.0931	-0.40
Juntos	MLP–ERM	59	0.2979	0.0931	-0.35
Juntos	MLP–equal	13	0.2968	0.0931	-0.23
Juntos	MLP–equal	31	0.2966	0.0929	-0.13
Juntos	MLP–equal	59	0.2968	0.0930	-0.18
Juntos	MLP–DRO	13	0.3064	0.0968	+4.51
Juntos	MLP–DRO	31	0.3062	0.0967	+4.38
Juntos	MLP–DRO	59	0.3057	0.0964	+4.48

For Pensión 65, the three neural ERM seed losses ranged from 0.3424 to 0.3426, compared with 0.3412 for the ensemble of predicted probabilities. The corresponding Juntos range was 0.2967–0.2979, compared with 0.2965 for the ensemble. Group DRO used the same three predetermined seeds under its revised selection protocol.

9 Validation design as a sensitivity analysis

Table S15. Alternative five-fold within-department approximate-cluster validation versus the outer LODO results. The protocols differ in model selection and training configuration; the difference is not a pure measure of geographic shift.

Task	Model	Cluster CV	LODO	Difference
P65	Source prevalence	0.5820	0.6228	+0.0408
P65	Logistic L2	0.3485	0.3551	+0.0066
P65	CatBoost	0.3161	0.3217	+0.0056
P65	MLP–ERM	0.3319	0.3412	+0.0093
P65	MLP–DRO	0.3404	0.3520	+0.0116
Juntos	Source prevalence	0.4210	0.4371	+0.0161
Juntos	Logistic L2	0.2953	0.3000	+0.0046
Juntos	CatBoost	0.2909	0.2962	+0.0052
Juntos	MLP–ERM	0.2915	0.2965	+0.0050
Juntos	MLP–DRO	0.2954	0.3050	+0.0096

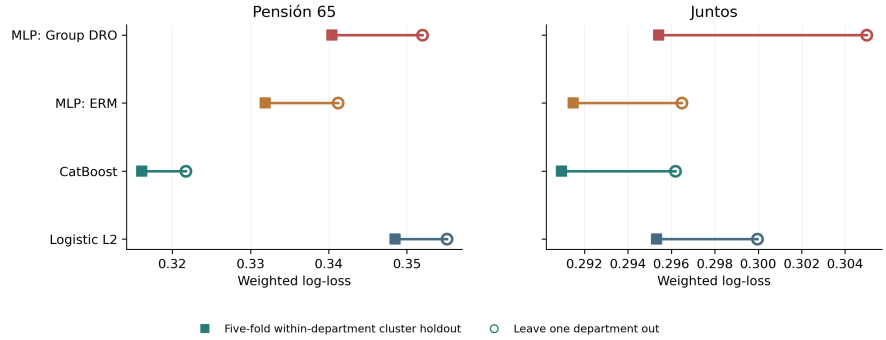


Fig. S5. Pooled log-loss under leave-one-department-out and within-department approximate-cluster evaluation; the fitted specifications and tuning procedures also differ.

CatBoost pooled log-loss under within-department conglomerate holdout was 0.3161 for Pensión 65 and 0.2909 for Juntos, compared with 0.3217 and 0.2962 under department exclusion. The former protocol used fixed configurations, whereas the latter included inner department-level model selection [11,4].

9.1 Exclusion of an entire natural region

Table S16. Natural-region exclusion experiment for the fixed CatBoost configuration. This trains on only two of the three natural regions and is not directly comparable with 24-source-department tuning.

Task	Held-out block	n	Log-loss	Brier	Bias (pp)	AUC
P65	Three held-out regions	11,354	0.3482	0.1103	+0.18	0.899
P65	Coast excluded.0	4,154	0.2566	0.0772	+7.68	0.864
P65	Highlands excluded.0	5,132	0.4293	0.1395	-8.16	0.895
P65	Amazon excluded.0	2,068	0.5790	0.1938	-12.81	0.828
Juntos	Three held-out regions	17,671	0.3101	0.0974	-1.59	0.854
Juntos	Coast excluded.0	6,230	0.1680	0.0440	+3.04	0.823
Juntos	Highlands excluded.0	6,651	0.4631	0.1567	-8.25	0.826
Juntos	Amazon excluded.0	4,790	0.4537	0.1474	-2.72	0.783

A fixed CatBoost specification was trained on two natural regions and evaluated on the third. Pooled losses were 0.3482 for Pensión 65 and 0.3101 for Juntos. The greatest loss was 0.5790 for Pensión 65 when Amazon was excluded and 0.4631 for Juntos when Highlands was excluded. This three-block experiment changes the source population more substantially than the 25-department analysis.

10 Measured source–target shift

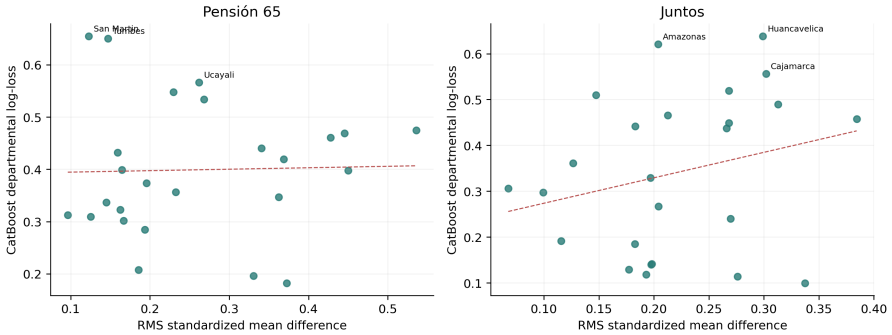


Fig. S6. Department-held-out CatBoost loss against the root-mean-square standardized difference of fourteen numeric predictors between the target department and its source departments. The diagnostic captures selected changes in covariate means, not all differences between distributions.

For each excluded department, covariate shift was summarized by the root-mean-square standardized difference in weighted means across 14 numeric features, including the training-fitted asset component. The Spearman correlation with CatBoost departmental loss was 0.115 for Pensión 65 and 0.240 for Juntos ($n = 25$ per task). This diagnostic measures shifts in selected means rather than the complete predictor distributions [6].

11 Specification and analytic-population sensitivity

Table S17. Fixed-CatBoost sensitivity analyses. Complete-case and all-household specifications use different analytic populations.

Task	Specification	n	Log-loss	Brier	AUC	Bias (pp)
P65	Fixed-spec baseline	11,354	0.3217	0.1006	0.914	+0.04
P65	Exclude asset component	11,354	0.3418	0.1074	0.901	+0.26
P65	Unweighted training	11,354	0.3240	0.1015	0.915	+1.44
P65	Add natural region	11,354	0.3098	0.0965	0.921	-0.05
P65	Complete-case baseline*	10,273	0.3129	0.0977	0.911	+0.19
P65	Complete-case + ethnicity*	10,273	0.3136	0.0979	0.911	+0.19
Juntos	Fixed-spec baseline	17,671	0.2961	0.0922	0.867	-0.22
Juntos	Exclude asset component	17,671	0.3071	0.0953	0.854	-0.20
Juntos	Unweighted training	17,671	0.2991	0.0935	0.866	-1.07
Juntos	Add natural region	17,671	0.2971	0.0920	0.866	-0.18
Juntos	Complete-case baseline*	16,824	0.2885	0.0895	0.869	-0.18
Juntos	Complete-case + ethnicity*	16,824	0.2888	0.0895	0.869	-0.16
Juntos	All households*	36,184	0.1384	0.0425	0.942	-0.07

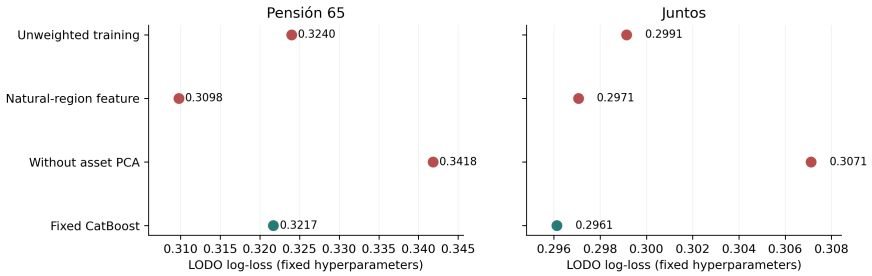


Fig. S7. Fixed CatBoost sensitivity analyses, including an otherwise identical specification without the fold-specific asset principal component. Population restrictions are tabulated separately.

With fixed CatBoost settings, log-loss was 0.3217 for Pensión 65 and 0.2961 for Juntos using the source-fitted asset component. Excluding that component gave 0.3418 and 0.3071, respectively. Adding natural region yielded 0.3098 and 0.2971, while omitting training survey factors yielded 0.3240 and 0.2991. These models use the same departmental evaluation partitions; the sensitivity estimates concern a fixed parameter configuration, not a repeated hyperparameter search.

Complete-case restriction reduced the Pensión 65 sample to 10,273 households and the Juntos sample to 16,824. Including indigenous identification within those complete-case samples changed loss from 0.31292 to 0.31363 and from 0.28846 to 0.28877, respectively. A further Juntos sensitivity used all 36,184 surveyed households rather than the age-restricted sample; its loss (0.1384) refers to a different prevalence and prediction population.

12 Alternative selection criteria and territorial reweighting

Table S18. Pooled and departmental predictive losses under alternative model-selection and source-department reweighting specifications. The two Group DRO models use the same architecture but different hyperparameter-selection protocols. All models are evaluated on held-out departments.

Task	Model	Pooled	Macro	Maximum	Max/H
P65	Logistic L2 (weighted)	0.3551	0.4382	0.6734	0.997
P65	Logistic L2 (equal departments)	0.3581	0.4333	0.6189	1.056
P65	Neural ERM	0.3412	0.4211	0.6715	1.047
P65	Neural equal departments	0.3414	0.4149	0.6374	1.004
P65	Group DRO (worst-group CV)	0.3520	0.4196	0.6161	1.113
P65	Group DRO (pooled CV)	0.3479	0.4168	0.6195	1.043
P65	CatBoost	0.3217	0.3990	0.6548	1.014
Juntos	Logistic L2 (weighted)	0.3000	0.3436	0.6498	1.336
Juntos	Logistic L2 (equal departments)	0.2992	0.3433	0.6384	1.356
Juntos	Neural ERM	0.2965	0.3414	0.6466	1.289
Juntos	Neural equal departments	0.2959	0.3411	0.6379	1.328
Juntos	Group DRO (worst-group CV)	0.3050	0.3528	0.6010	1.695
Juntos	Group DRO (pooled CV)	0.2978	0.3424	0.6151	1.473
Juntos	CatBoost	0.2962	0.3402	0.6384	1.288

Table S19. Exploratory paired differences in pooled and maximum departmental log-loss under alternative objectives. Negative values indicate smaller loss for the first method. Maximum-loss intervals re-identify the maximizing department in each of 1,600 within-department approximate-cluster resamples.

Task	Comparison	Pooled Δ	Maximum Δ	95% interval (maximum)
P65	Worst vs pooled DRO	+0.0041	-0.0034	[-0.0370, +0.0401]
P65	Balanced vs standard LR	+0.0030	-0.0545	[-0.0785, -0.0242]
P65	Worst-selected DRO vs ERM	+0.0108	-0.0554	[-0.1537, +0.0100]
Juntos	Worst vs pooled DRO	+0.0072	-0.0140	[-0.0296, -0.0045]
Juntos	Balanced vs standard LR	-0.0008	-0.0114	[-0.0192, +0.0062]
Juntos	Worst-selected DRO vs ERM	+0.0085	-0.0455	[-0.0677, -0.0263]

Table S20. Frequency of selected Group DRO hyperparameters across 25 outer training partitions. The statistic minimized during inner selection is the maximum weighted loss among source validation departments. The last column reports the median of this criterion among folds selecting a given combination.

Task	Weight decay	DRO step	Folds	Median worst validation loss
P65	0.002	0.04	7	0.711
P65	0.002	0.12	5	0.659
P65	0.020	0.04	7	0.649
P65	0.020	0.12	6	0.645
Juntos	0.002	0.04	0	–
Juntos	0.002	0.12	13	0.626
Juntos	0.020	0.04	1	0.609
Juntos	0.020	0.12	11	0.606

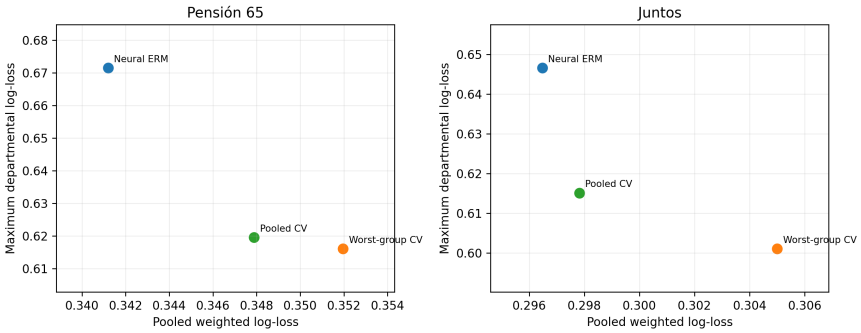


Fig. S8. Two Group DRO validation-selection protocols and neural ERM evaluated using the same departmental holdouts. Worst-department and pooled scores correspond to the out-of-department predictions, not the source-validation criterion used for tuning.

Group DRO was refitted under two inner-validation selection criteria: pooled household-weighted loss and maximum department-specific loss. The latter selected both weight decay and update step, whereas the former reused the ERM-selected weight decay. Their differences therefore reflect the specified joint selection protocols, rather than a change in validation data. The geographically balanced

logistic model used weights scaled to give each source department the same total weight; the logistic penalty was inherited from the household-weighted logistic fit. All these predictions were obtained from complete department holdouts, with the asset component recalculated within each source partition. The two alternative optimizations and the equal-department neural specification offer different forms of territorial reweighting.

13 Methodological scope

The main analysis uses two age-restricted samples from one survey year. Official survey-stratum identifiers were not included in the analytic dataset; the approximate-cluster intervals therefore omit full stratification and fitted-model variability [5,10]. The fixed-configuration and all-household sensitivity analyses differ from the primary tuning protocol and population, respectively.

The replication materials include data-construction code, grouped predictor transformations, fold-specific component estimates, held-out predictions and evaluation scripts. Analyses in additional survey years would allow assessment of temporal transport alongside geographic variation.

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